

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Choosing the Right Developing Model



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Give detailed information related to UML and common related topics.

Can Do Statements:

- I can read about XP and UML.
- I can write a text to summarize ideas related to XP and UML.
- I can give details about XP and UML in a written text.

Ask your partner:

- What steps should we take into account when designing software?
- Do you think it is possible to mix and use different developing models interchangeably? Why?



**Let's Explore!**

Jeffrey is reading a blog about modeling approaches in order to make a summary for her final presentation.

UML & XP

Instructions: Read the text below.

As a software analyst, one of the many functions you can have among your duties is to diagram and sketch how a program has to work. To do so, it is important to know what the program or application has to do, and what individuals or areas of the company are responsible for it to work.

During the planning stage, it is important to define what modeling approach you are using to sketch the workflow of the project. Among the most common ones to work with, you can opt to choose the Unified Modeling Language (UML), which allows you to diagram and model the way the software creation process should be carried out, as it diagrams the way the workflow correlates one dependency to another. Or you could always opt to choose the eXtreme Programming approach (XP), which offers a more relaxed schedule for programmers, stakeholders and individuals to accomplish their goals, as the commonly end-of-day practice can be changed or modified to an end-of-week assignment.

Same way, some people can agree with the fact that both approaches are essential when designing software products as they both can relate to AM (Agile Modeling) principles; however, some others consider it is advisable not to combine them because they are both alike in some aspects. Such claims are based on the fact that UML and XP have similar tools, such as use cases and user stories; therefore, these tools aren't able to perform the same actions as XP seeks to segment tasks in order to ease processes while improving quality.

As an overall, you must have clear what you are aiming your software to do in order to use either XP or UML as a modeling approach.



Let's Practice!

XP and UML approaches are both useful.

Instructions: Read the question. Select all the correct answers.

1. Based on the information from the blog, which of the following statements about XP and UML are true?
 - a. Both approaches are essential when designing software products.
 - b. Both modeling approaches are used to sketch the workflow of software projects.
 - c. When sketching a program, you can combine both approaches.
 - d. They both can relate to AM principles.
 - e. They both offer a relaxed schedule for programmers.
 - f. They are both alike because they offer similar tools.



Let's Practice!

How can the software creation process be carried out?

Instructions: Match column A with column B. Read the sentences aloud. Check your pronunciation with your instructor.

COLUMN A

1. During the planning stage, it is important to define...
2. Among the most common ones to work with,...
3. UML allows you to diagram and model...
4. Or you could always choose the eXtreme Programming approach (XP), ...
5. Use cases and user stories...

COLUMN B

- a. you can opt to choose the Unified Modeling Language (UML).
- b. what modeling approach you are using to sketch the workflow of the project.
- c. able to perform the same functions.
- d. which offers a more relaxed schedule for programmers.
- e. the way the software creation process should be carried out.



Extension Activity

Similarities and Differences between UML and XP

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about UML and XP.
- Create a comparative chart in which you include the most relevant similarities and differences between both modeling approaches.
- Make a poster with your chart.
- Show it to your classmates and instructor.

Self-Assessment

Yes

No

- I look for information about UML and XP.
- A create a comparative chart with the similarities and differences between both modeling approaches.
- A make a poster with a comparative chart.

☐ ☐
☐ ☐
☐ ☐


Target Words

- **sketch** (verb): to make a provisional drawing about something
e.g. One of the many functions you can have among your duties is to diagram and sketch how a program has to work.
- **opt** (verb): to make a decision over something
e.g. You can opt to choose the Unified Modeling Language.
- **correlate** (verb): to link 2 things; to make a relation between two objects
e.g. It diagrams the way the workflow correlates one dependency to another.
- **stakeholder** (noun): an individual or a group of people involved with a project
e.g. You could always opt to choose the extreme programming approach (XP), which offers a more relaxed schedule for programmers, stakeholders and individuals to accomplish their goals.
- **AM** (acronym): Agile Modeling; an approach to software modeling
e.g. Both can relate to AM principles.



Learning Points

Both as an Adjective

The word **both** has different grammatical functions in English, one of the most common is its use as an **adjective**.

As an *adjective*, **both** replaces two objects before a noun and it has the meaning of *one and the other* or *two together*.

e.g.

*The company had to design two different programs today, **both** programs are important to make the company work.*

Both tools are key in software developing.

It's important to note that **both** can also be either a pronoun or a conjunction. Let's check the differences:

As a *pronoun* **both** replaces two nouns.

e.g.

*The company hired two different companies to develop the software; **both** were really good options.*

As a *correlative conjunction*, **both... and...**, the elements that are after **both** and **and** must be parallel, that means they must have some relation in terms of balance.

e.g.

*She is **both** a programmer **and** analyst.*

*The company determined **both** to analyze the market and to develop **and** app for the company.*

Examples

- *Same way, some people can agree with the fact that both approaches are essential when designing software products as they both can relate to AM (Agile Modeling) principles.*
- *Some others consider it is advisable not to combine them because they are both alike in some aspects.*



Learning Points

Tenses Review

Modal auxiliary verbs, simple and continuous tenses have some similarities and differences, let's compare them.

Among the modal auxiliary verbs, the most common ones are those expressing ability and/or probability, for example **can** and **could**.

When they express probabilities, they can be used interchangeably.

E.g.

*This program **can** run in any OS, just give it a try.*

*This program **could** run in any OS, just give it a try.*

When they are used to talk about abilities, **could** is the past form of **can**.

E.g.

*This diagram **can** help you organize your ideas. (you can use it)*

*That diagram **could** help you organize your ideas. (you didn't use it)*

Be able is a similar expression that means **can** when it expresses ability.

E.g.

*This diagram **is able to** help you organize your ideas.*

*This diagram **was able to** help you organize your ideas.*

Simple Tenses

Simple tenses can be in either present, future or past, and commonly use the following structure:

Subject + (aux) + verb + complement.

Use simple tenses to express general ideas and facts.

In present form, you use the auxiliary **do** to make questions and express negative statements.

E.g.

*You **need** to install that app.*

*You **don't need** to install that app.*

***Do** you **need** to install that app?*

In past form, you use the auxiliary **did** to make negative statements and questions.

E.g.

*You **needed** to install that app.*

*You **didn't need** to install that app.*

***Did** you **need** to install that app?*





Learning Points

In future form, you use the modal auxiliary verb **will** to make affirmative and negative statements and questions.

E.g.

You **will need** to install this app.

You **won't need** to install this app.

Will you **need** to install this app?

Continuous Tenses

You can use tenses in continuous form to express ongoing actions in either present, past or future times.

Continuous forms are characterized by the use of the verb **to be** as auxiliary verb and the prefix **-ing** after the main verb.

Structure:

Subject + to be + verb + ing + complement.

Present tense:

E.g.

I'm currently running a test on this program, to see if it works.

Are you **running** a test on this terminal?

I'm not running a test on that computer right now, you can use it.

Past tense:

E.g.

I was sketching the designing of our next project.

I wasn't thinking on that matter when I made the design.

Weren't you **working** on that project with Alisson?

Future tense:

Future time is commonly used with the auxiliary **will + ing** forms.

E.g.

I will be working on that design the whole day tomorrow.

I won't be coming to the office tomorrow because I'll be designing the web page.

Will you **be working** on the project tomorrow?





Learning Points

Examples

- *As a software analyst, one of the many functions you can have among your duties is to diagram and sketch how a program has to work.*
- *During the planning stage, it is important to define what modeling approach you are using to sketch the workflow of the project.*
- *Therefore, these tools aren't able to perform the same actions as XP seeks to segment tasks in order to ease processes while improving quality.*

Sentence Variety

Sentence variety allows your texts to have rhythm and flow, which makes easier for readers to understand your ideas.

There are multiple techniques to create sentence variety in your texts, these are some of them:

1. You can start some of your sentences by an *adverb*.

Example:

Frequently, software developers use UML since it includes meta object facility and graphic notation.

2. You can join your ideas using a *verb ending in ING*.

Example:

Two sentences

UML uses object-oriented programming. It allows programmers to create visual models of systems.

Joined with ING form

UML allows programmers to create visual models of systems **using** object-oriented programming.

3. You can join your ideas using a *verb in past participle*.

Example:

Two sentences

UML executable could run entire applications. They were deployed in different types of platforms.

Joined with a past participle

Deployed in different types of applications, UML executables could run entire applications.